Grounded-input noise

PASS - noise magnitude below the 25 µV target. Measured standard deviation was 18.741 µV. The three measured peak bands account for only 0.21% of total noise power. They contribute little to the overall noise.

Setup and measurement

ADS127L18 board: CH0P and CH0N each shorted to analog ground; generator disconnected; Feather DAC disabled. Normal ADC-to-ring-buffer-to-SD recording remained active. Recorded 29.999 s; analyzed 1,024,000 individual SD readings from 1.000 to 2.000 s. Sample rate: 1,024,000/s. Voltage conversion assumes the nominal 2.5 V reference.

Statistic	Measured value
Mean offset from ground	+49.544 µV
Standard deviation	18.741 µV
Minimum / maximum	-44.703 / +151.992 µV
Peak-to-peak range	196.695 µV

RMS noise contributions

Frequency band	RMS noise
120 Hz ±3 Hz	0.685 µV
240 Hz ±3 Hz	0.464 µV
311,147 Hz ±3 Hz	0.260 µV
All remaining frequencies	18.711 µV
Total from FFT	18.731 µV

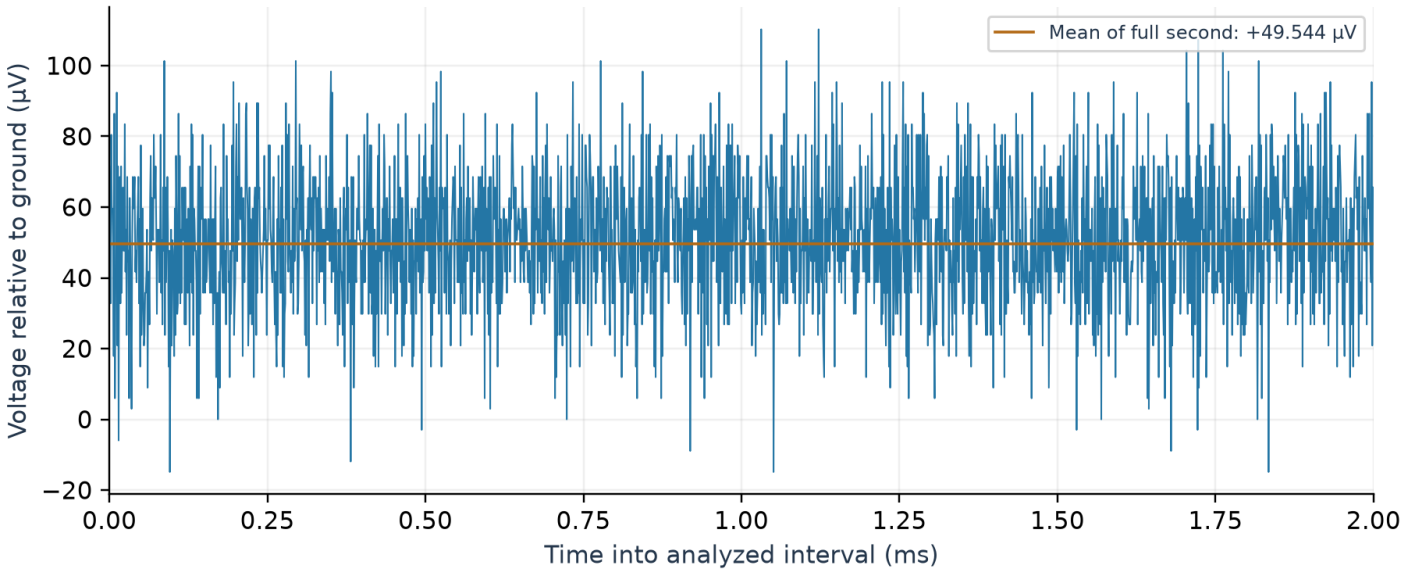
RMS contributions combine by squaring, adding and taking the square root. Each listed band includes both the peak and background noise within ±3 Hz. Other frequencies, including the smaller peak near 360 Hz, remain in the remainder. The FFT total differs slightly from standard deviation because the FFT uses a Hann window.

Recording integrity and conclusion

92,158,293 sample bytes saved; one file change; no reported receiver errors or ring overflow. Retained-file checksums matched. The observed noise meets the requested magnitude target. The small repeating components are documented; the remaining noise has not been proven random.

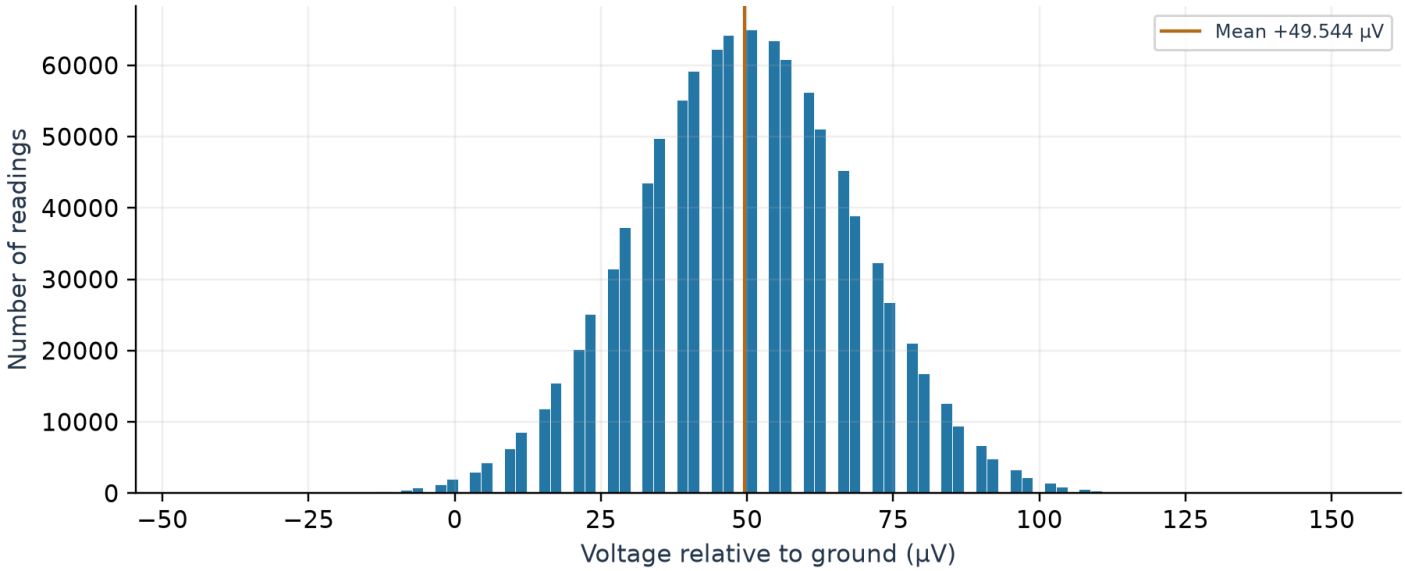
Readings and distribution

1. Individual readings



Every reading in the first 2 ms of the analyzed second is shown, without averaging. The horizontal line marks the mean of the full second. This short view does not show every extreme found across all one million readings.

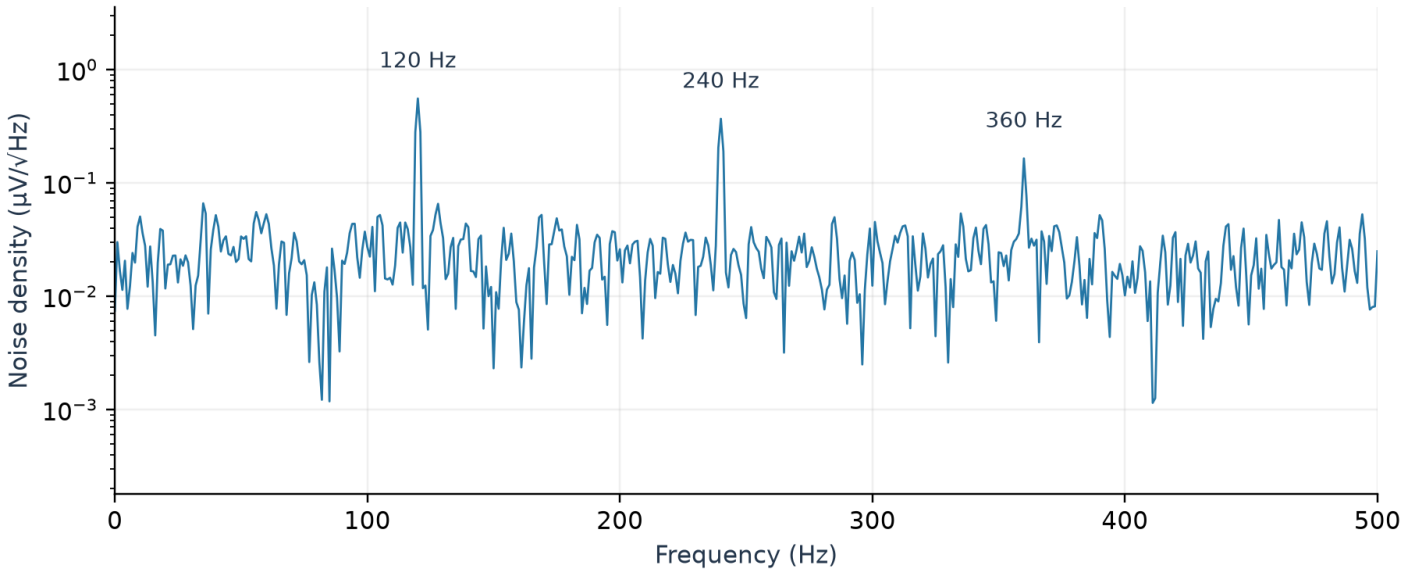
2. Distribution of all readings



Approximately bell-shaped and centered at +49.544 µV. The mean is an offset from expected zero; the 18.741 µV standard deviation measures the spread around that mean. A bell-shaped distribution alone does not establish randomness or independence between readings.

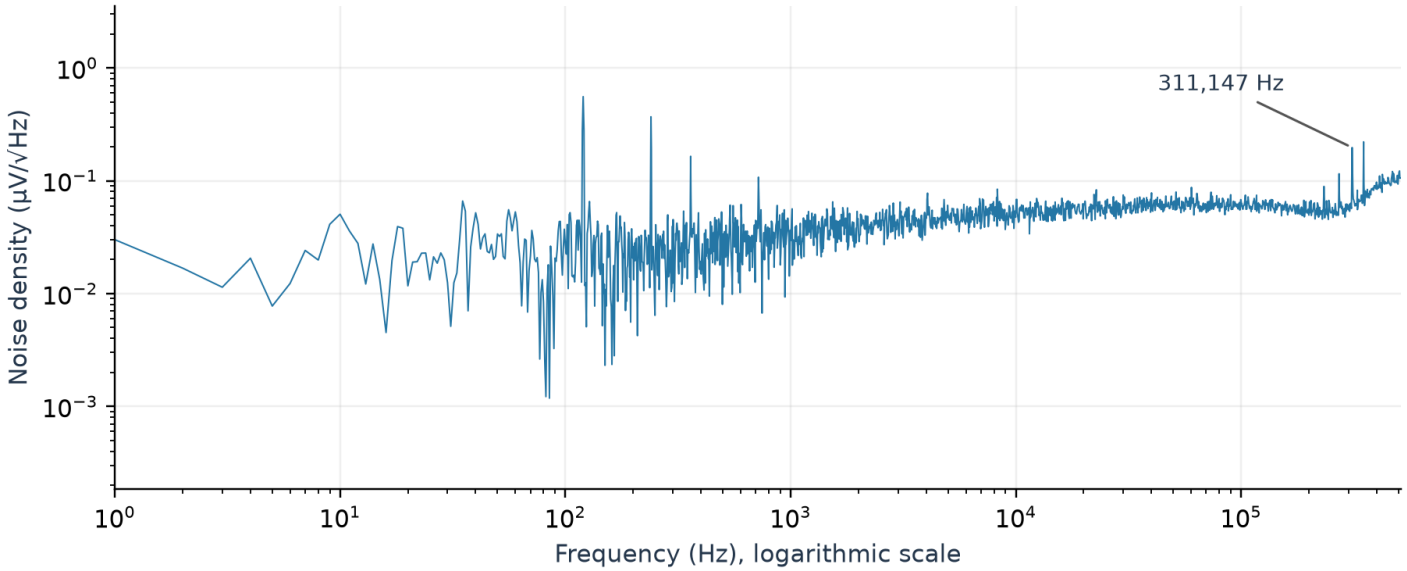
Noise by frequency

3. Low-frequency spectrum



The 120 Hz peak and its multiples at 240 and near 360 Hz suggest a repeating disturbance. Its source is not identified. The vertical scale is logarithmic. Peak height is noise density, not the RMS contribution; that requires combining power across the peak width.

4. Full frequency range



Mean removed; periodic Hann window; no FFT averaging; 1 Hz bins and 1.5 Hz window noise bandwidth. Full-range display retains the strongest bin in each plot group. ADC filtering can shape random noise. **Source:** captures/hardware-smoke/20261031-170006/ - noise-samples.bin, result.json, feather.txt and p4.txt. This report characterizes one analyzed second and the assembled board, not the ADC alone.